

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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Total Marks: 100

Examination Paper

Question Q1 (20 marks)

- a) List the three main types of integrity constraints in relational databases and provide a brief explanation for each. (2 marks) (2)
- b) Design an EER diagram to represent a university's student database, including entities such as Student, Course, and Professor. Ensure that the diagram accurately reflects the relationships between these entities. (4 marks) (4)
- c) Explain the concept of normalization in relational databases and provide an example of a normalized schema for a university's student database. Be sure to discuss the benefits of normalization. (3 marks) (3)
- d) Create a stored procedure that updates the mileage of a vehicle assigned to a booking and adds a record to the payment table when the distance is completed. Assume that the Booking table is updated only when the distance is assigned at the completion of a booking. Provide the SQL code for this procedure, including any necessary parameters and constraints. (5 marks) (5)
- e) Design a relational schema to store information about university courses, including course names, descriptions, credits, and instructors. Ensure that the schema is normalized and includes all necessary tables and relationships. Provide a brief explanation for your design choices. (6 marks) (6)

Question Q2 (20 marks)

- a) What is the normal form of the relation in Figure 22? (3)
- b) If the relation is not in 2NF, normalize it to 2NF by removing any non-prime attributes that are transitively dependent on any key. (2)
- c) Is the relation in Figure 23 in 3rd Normal Form (3NF)? Explain your answer. (1)
- d) If the relation is not in 3NF, normalize it to 3NF by removing any non-prime attributes that are transitively dependent on any key. (2)
- e) Consider a relation R with functional dependencies $F = \{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. Compute the keys for relation R. (2)
- f) What is the difference between a superkey and a key in the context of relational

databases? (2)

g) Explain the concept of Boyce-Codd Normal Form (BCNF) and how it is used to normalize relations. (2)

h) Design an ER diagram for a university's library system, including entities such as 'Book', 'Author', and 'Section'. (2)

i) Map the following ER diagram to a relational schema: [FIGURE: slide_013_fig_1.png]. Explain your design choices. (4)

Question Q3 (22 marks)

a) List the three main types of integrity constraints in a relational database system. (2)

b) Explain the concept of specialization and generalization in the Entity-Relationship (ER) model, providing an example from the given scenario. (2)

c) Design a relational schema to represent the 'Member' entity in the library system, including primary key, foreign keys, and attributes. (2)

d) Construct an EER diagram for the given scenario, representing the relationships between entities and their attributes. (3)

e) Explain how to map an N-ary relationship in the ER model to a relational schema, providing an example from the given scenario. (2)

f) Design a trigger to ensure that no customer withdraws more than the given limit (40000) per day in the bank's database system. (3)

g) Explain how to handle overlapping constraints between entity types in the ER model, providing an example from the given scenario. (2)

h) Map the 'Member' entity to a relational schema, including primary key, foreign keys, and attributes, considering the given constraints. (3)

i) Design an assertion to ensure that no employee has a salary greater than their manager's salary in the database system. (3)

Question Q4 (20 marks)

a) Explain the process of logical database design and its importance in database development. (3)

- b) Design a conceptual ER model for a university's student information system, including entities such as Student, Course, and Department. (3)
- c) Explain the concept of normalization in database design and provide an example of a normalized schema. (1)
- d) Given a conceptual ER model, identify the entities that are dependent on another entity and explain how to create a composite key. (3)
- e) Design a schema for a university's course registration system, including tables for courses, students, and registrations. (3)
- f) Explain the process of converting an ER model to a relational model, including entity-relationship diagrams and data normalization. (4)
- g) Given a relational schema, identify the primary key and foreign keys, and explain how to create indexes for efficient query performance. (3)

Question Q5 (18 marks)

- a) List the three main types of integrity constraints used in EER models. (5)
- b) Consider a university with several departments. Each department has a name, and each employee is assigned to one department. Create an EER diagram to represent this scenario. (2)
- c) Explain the concept of specialization in EER models and provide an example. (2)
- d) An employee can be assigned to one or more departments. If an employee is assigned to both a department and a project, which attribute should be used as the primary key for the employee's record? (2)
- e) List the four main options for mapping ISA relationships in EER models. (1)
- f) Consider a student who is enrolled in both a degree program and a certificate course. How would you represent this relationship using an EER diagram? (1)
- g) Explain the concept of referential integrity and provide an example. (1)
- h) Consider a university with several departments. Each department has a budget, and each employee is assigned to one department. Create an EER diagram to represent this scenario. (2)
- i) Explain the concept of data independence in database systems. (2)